ingredients

① data points (known)

$(\vec{x}_1, y_1), \ldots, (\vec{x}_n, y_n)$

$(x_{i1}, \ldots, x_{id}) \in \mathbb{R}^d$

② model / relationship

$y = f(\vec{x}) + \varepsilon$

model — noise

↳ parametric

↳ linear

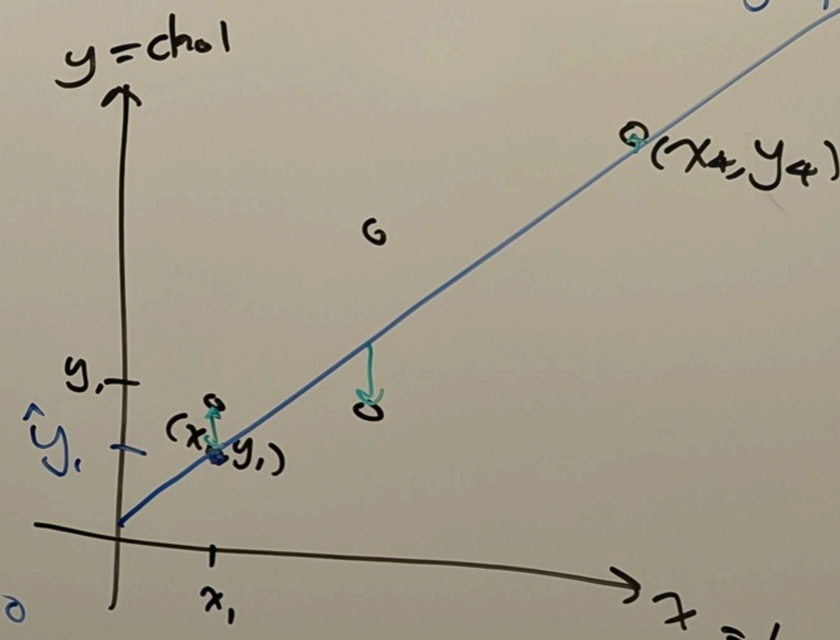
$y = \beta_0 + \beta_1 x_1 + \ldots + \beta_d x_d + \varepsilon$

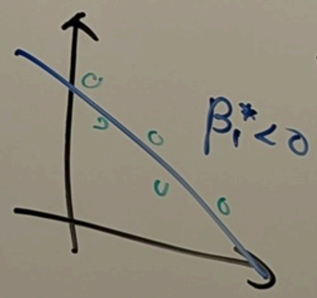
intercept — slopes

③ estimation / pick best parameters

$\beta_0, \beta_1, \ldots, \beta_d$? ? ?

$y = \beta_0 + \beta_1 x$

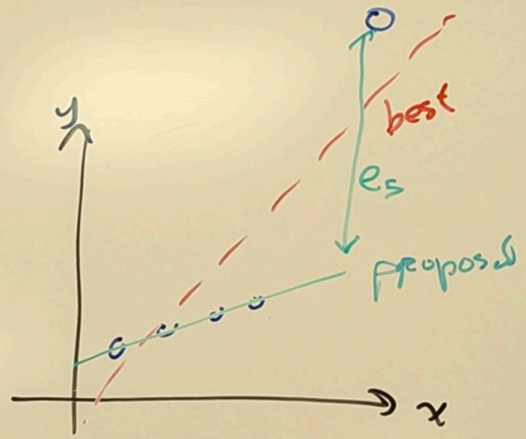




$\beta_1^* = \text{Corr}(x,y) \cdot \frac{SD(y)}{SD(x)}$ (both > 0)

Corr(x,y): sample correlation

④ metric / criterion / objective / loss function



① more flexible model

② throw away + sensitivity

$$\text{mean squared error} = \frac{1}{n}\sum_{i=1}^{n}(r_i)^2$$ ← squared

$$= \frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2$$ ← "fitted value"

"MSE"

"predicted" true

$r_i = y_i - \hat{y}_i$ ← residual / error

e_i

< 0 over-estimate

$$MSE = \frac{1}{n}\sum_{i=1}^{n}((\beta_0 + \beta_1 x_i) - y_i)^2$$

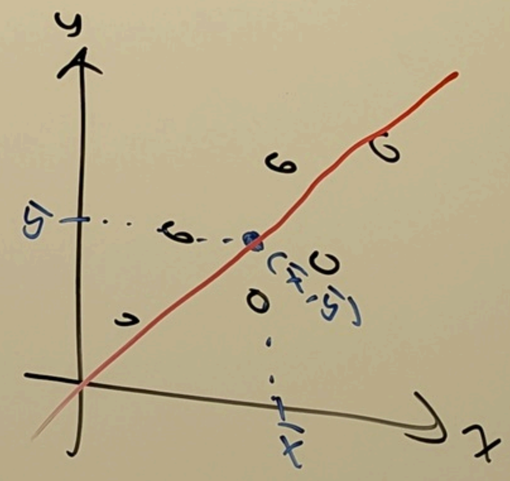
x_i, y_i: known

$$\beta_0^*, \beta_1^* = \min_{\beta_0, \beta_1}(\ldots)$$

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

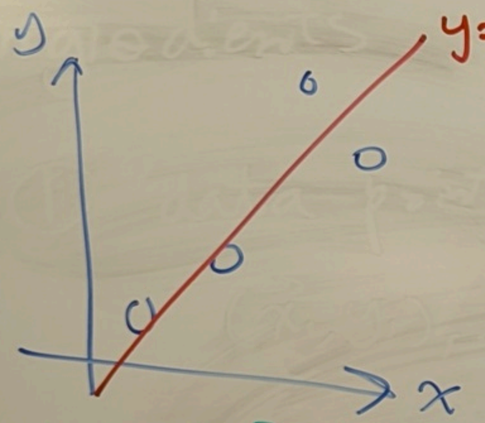
$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$$

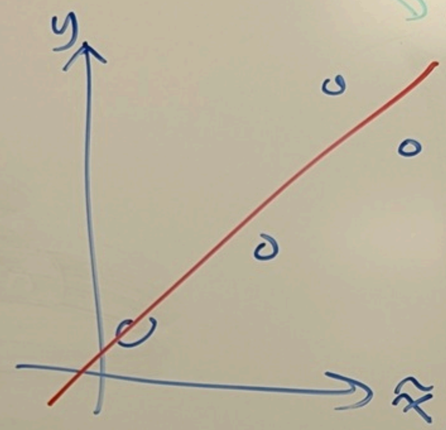
$$\bar{y} = \beta_0^* + \beta_1^* \bar{x}$$

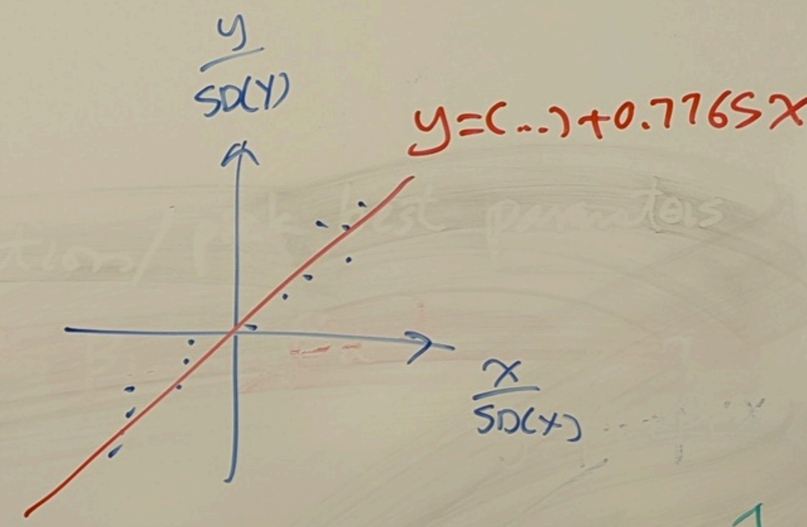


β_0^* β_1^*

$y = 3 + 5x$ $=$ $y = 3 + 05\tilde{x}$







$\beta_1^* = corr(x,y) \frac{\cancel{SD(x)}}{\cancel{SD(y)}} \frac{1}{1}$

linear transformation

$10x$

x^2	x	y

$\tilde{x}$	y

$\tilde{\tilde{x}}$	y

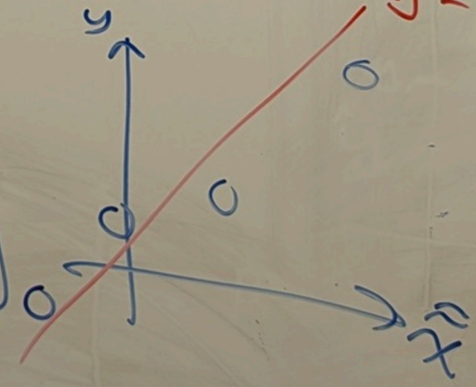
$(y \sim x + x^2)$

$x - 2$

$x \longmapsto c + sx$

$1 \quad 11$

$0 \quad 10$

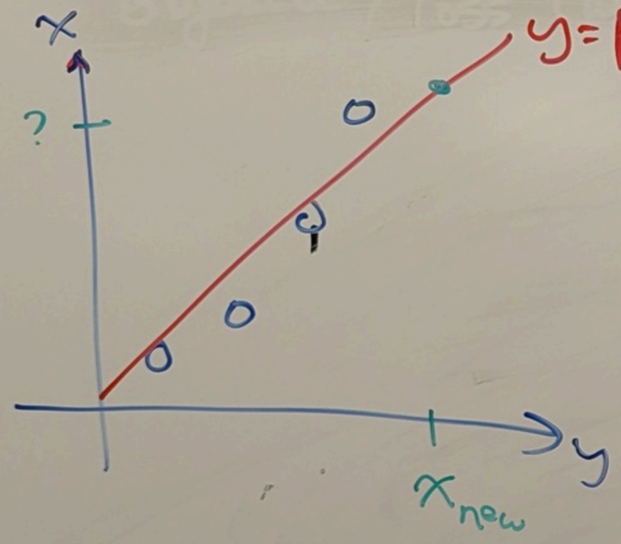


$y = 13 + 5\tilde{\tilde{x}} = 3 + 5x$

$R: \overset{scale()}{x \longmapsto} \frac{x - \bar{x}}{SD(x)}$

$\left(-\frac{\bar{x}}{SD(x)}\right) + \left(\frac{1}{SD(x)}\right) x$

c s



$y = \beta_0^* + \beta_1^* x$

$$\sum_{i=1}^{n} \left[y_i - (\beta_0 + \beta_1 x_i) \right]^2$$

$$\|\vec{v}\|^2 = \sum_{i=1}^{d} (v_i)^2$$

$$\left\| \vec{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} - \left[\beta_0 \vec{1} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} + \beta_1 \vec{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \right] \right\|^2$$

$$= \left\| \begin{pmatrix} y_1 - (\beta_0 + \beta_1 x_1) \\ \vdots \\ y_n - (\beta_0 + \beta_1 x_n) \end{pmatrix} \right\|^2$$

df

x	y

R: lm(df$y ~ df$x)

lm(y ~ x, data = df)

MLR

parameters

$$y = \beta_0 + \beta_1 x_1 + \ldots + \beta_d x_d + \varepsilon$$

predictors

data frame

y | $x_1 \ldots\ldots x_d$

y_1 | $x_{11} \ldots\ldots x_{1d}$

$\vdots$ | $\vdots$

y_n | $x_{n1} \ldots\ldots x_{nd}$

n × 1 | n × d

"design" matrix

← same →

$$\begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \beta_0 \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} + \begin{bmatrix} & X & \end{bmatrix} \times \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_d \end{bmatrix}$$

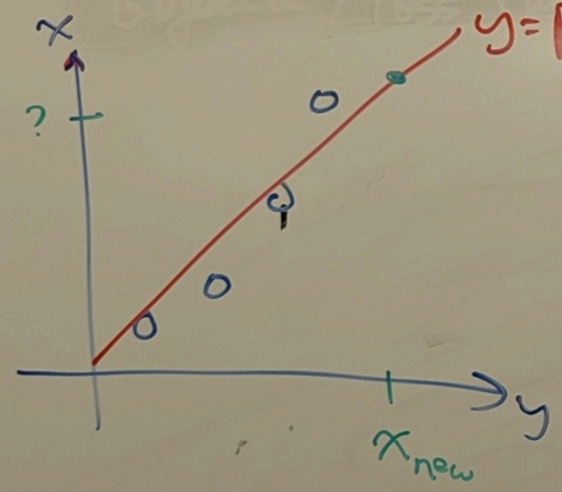
matrix mult.

$$= \underbrace{\begin{bmatrix} 1 & x_{11} \ldots\ldots x_{1d} \\ \vdots & \vdots \\ 1 & x_{n1} \ldots\ldots x_{nd} \end{bmatrix}}_{n \times (1+d)} \times \underbrace{\begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_d \end{bmatrix}}_{\beta}$$

"constant" feature

$$\left\| \underbrace{\begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}}_{\text{output}} - \begin{bmatrix} 1 & X \end{bmatrix} \times \beta \right\|^2$$

Back 7:40pm



$y = \beta_0^* + \beta_1^* x$

$$\sum_{i=1}^{n} \left[y_i - (\beta_0 + \beta_1 x_i) \right]^2$$

$$\|\vec{v}\|^2 = \sum_{i=1}^{d} (v_i)^2$$

$$\left\| \vec{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} - \left[\beta_0 \vec{1} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} + \beta_1 \vec{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \right] \right\|^2$$

$$= \left\| \begin{pmatrix} y_1 - (\beta_0 + \beta_1 x_1) \\ \vdots \\ y_n - (\beta_0 + \beta_1 x_n) \end{pmatrix} \right\|^2$$

$$\vec{\beta}_* = (X^T X)^{-1} (X^T) \vec{y}$$

df

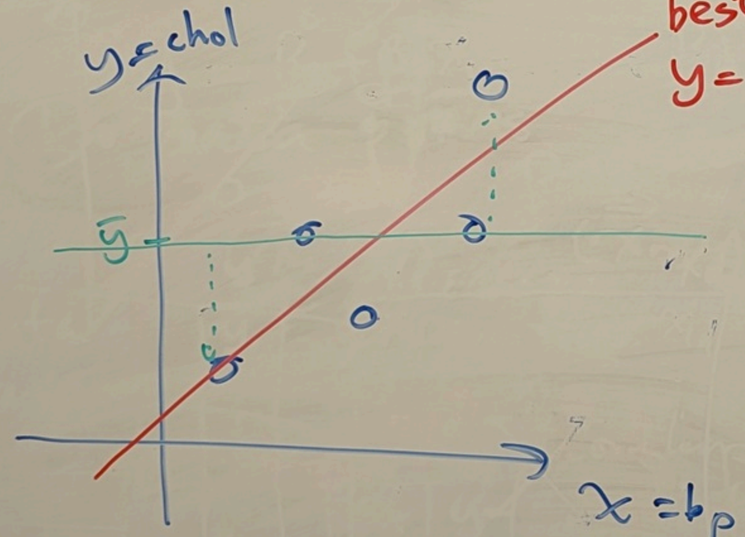


R: lm(df$y ~ df$x)

lm(y ~ x, data = df)

R^2

$$\vec{\beta}^* = (X^T X)^{-1} X^T \vec{y}$$



best lm

$y = \beta_0^* + \beta_1^* x_1 + \ldots + \beta_d^* x_d + \ldots$

best constant

$y = \beta_0^* = \bar{y}$

SS_{total} $\quad$ SS_{error} $\quad$ SS_{reg}

$$\sum_{i=1}^{n} (y_i - \bar{y})^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2$$

↑ small ↑ large

Constant model $>$ your model

fixed Constant

e.g. 10.35

do well

$y_i \approx \hat{y}_i$ so $\hat{y}_i \neq \bar{y}$

$$1 = \frac{SS_{error}}{SS_{total}} + \frac{SS_{reg}}{SS_{total}}$$

R^2

$$R^2 = \frac{\sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

← variation explained by model

total variation of data point

$$= 1 - \frac{SS_{error}}{SS_{total}}$$

$$0 \le R^2 \le 1$$

Constant model

$R^2 = 0.87$

perfect fit

no error

$R^2 = 1$

$corr(y, \hat{y}) = 1$

$y = a + b\hat{y}$

$R^2 \approx 0$

$$SS_{error} = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \approx SS_{total} = \sum_{i=1}^{n}(y_i - \bar{y})^2$$

Constant model

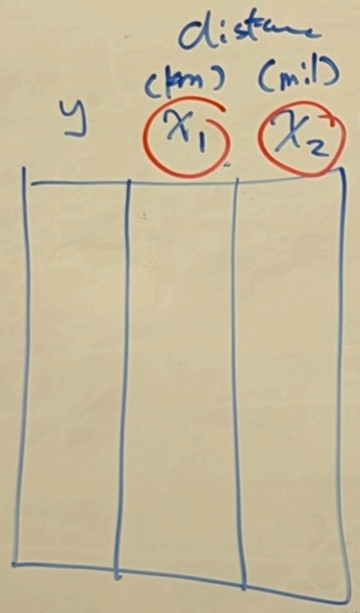
$$SS_{error} = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = 0$$

no mistake

$$corr\left(\begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}, \begin{bmatrix} \hat{y}_1 \\ \vdots \\ \hat{y}_n \end{bmatrix}\right)^2 = R^2$$

$corr(y, \hat{y}) = 0.8$ $R^2 = 0.8^2 = 0.64$

data



fitted model

$y = 1 + 20x_1$

$x_2 = 10x_1$

$corr(x_1, x_2) = 1$

$lm(y \sim x_2)$ $\quad y = 1 + 2x_2$

$lm(y \sim x_1 + x_2)$ $\quad y = 1 + ?x_1 + ?x_2$

$= 1 + \underline{20}x_1 + \underline{0}x_2$

$= 1 + \underline{0}x_1 + \underline{2}x_2$

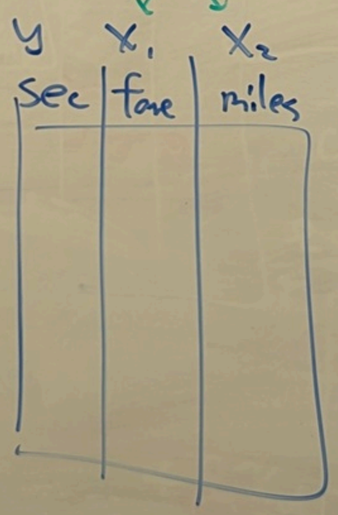
$= 1 + \underline{10}x_1 + \underline{2}x_2$

$lm(y \sim x_1)$

$lm(y \sim x_2)$

$y \sim x_1 + x_2$

corr ≈ 1



$lm(sec \sim fare + miles)$

$= +10^6 \quad -10^6$

$lm(y \sim x_1)$ $\quad y = 1.5 + 2.57x_1$

$lm(y \sim x_1 + x_2)$ $\quad y = (..) + 5935x_1 - 5932x_2$

$\approx x_1$

0.47

y	x_1	x_2	x_3
		-0.57	0.44

$x_3 = 1.2 + 3x_1 + 5x_2$

$?x_1 + ?x_2 \approx x_3$ $\quad lm(x_3 \sim x_1 + x_2) \rightarrow$ good fit $\rightarrow R^2 = 0.9$ large